Chapter 2

Literature Review

2.1 Background

Social sounds are emitted by both male and humpbacks, but songs are made exclusively by males (Pack et al., 2005; Payne and McVay, 1971). Songs serve a variety of purposes. They are most commonly used by lone whales to attract mates. According to Payne et al. (1983), a humpback whale song unit is “the shortest sound in the song which seem continuous to the human ear”.

Humpback whale songs vary based on geographic location, season, and at the individual level (Au et al. 2006). They identify at least nine distinct units of humpback whale song during the 2022 winter season in Hawaii. Some songs include a flat tonal groan, a vibrating upsweep, and a double upsweep. The number of units and the order to which units were sung varied considerably. General sequences of song units were repeatedly identified, producing a theme.

Allen et al. (2018) took a closer look at song level variation and theme level variation. Vocalization units in a sequence were grouped into a phrase. Repeated phrases create a theme. Data were collected from 2002 to 2014 on eastern Australian humpback whales. They generated ‘complexity scores’ based on observed aspects of theme and song. Their results showed evidence of an increasing number of sound units, unit types, and themes as song complexity increased. They believe that song complexity has been increasing over time due to evolving themes. As songs become more complex, they become more unique.

The first known instance of recorded Humpback whale sound was recorded in Bermuda in 1952 (Payne & McVay, 1971). Some of these sounds were verified as Humpbacks by their white flippers on their occasional breach of the water surface. However, many of these recorded sounds could not be verified with the technology of the time.

With the technology of today, higher quality sound recordings can be obtained, and visualized. This, in addition to additional knowledge about humpback whales, can be used to more easily verify sounds as humpback whales. The use of artificial intelligence to bypass the need for any human inspection to properly identify different whale (and other animal) calls is relatively novel.

Despite the advancements in artificial intelligence, humpback whales are still a challenge to detect. The high variability of humpback whale songs means that models must be highly adaptable. These higher adaptability models means that determining the key elements of a song is difficult.

This section explores the preprocessing methods, model implemented, and results of several different analyses involving signal detection and classification. Some reviewed literature focuses on other whale species such as fin, beluga, and sperm whales. The literature discussed spans the world, including the North Atlantic, the North Pacific, and the Caribbean. Non-whale literature focuses on birds and even non-animal such as deepfake audio detection.

Preprocessing steps vary widely between studies. Some studies perform direct sound classification, while others convert sound data to a spectrogram, then perform image classification.

An important note is the difference between audio detection and audio classification. The goal of audio detection is to pick out sections of interest, given an extended continuous audio file that contains more than one unit. The goal of audio classification is to identify the type of sound when provided with a short snippet of sound that contains a singular unit. Audio classification can more easily be designed to predict multiple classes. Bird species audio classification is one such example. We focus on audio detection, but use some information and knowledge gathered from audio classification to support our methods.

2.1 Humpback Whale Studies

Allen et al. (2021) focused on thirteen sites throughout the North Pacific (cite). These sites provided the researchers with over 187,000 hours of data, spanning over 14 years. They use a convolutional neural network and spectrogram image classification. They were able to obtain a model that worked well across the different locations. ROC-AUC scores were consistently 0.98 and larger, precision scores were all above 0.85, and recall scores were all above 0.82, with many above 0.90.

Kather et al. (2024) used annotated humpback whale song data on 60,000 audio segments from North Atlantic whales, each 4 seconds in length (cite). This analysis builds upon Allen et al.'s (2021) study, by performing additional preprocessing steps to ensure accuracy in higher variability scenarios. They found that their best model utilized all three preprocessing steps. Recall, precision, and F-scores of this model were 0.8, 0.87, and 0.84, respectively.

The other sources of noise and their frequencies play an important role in the machine learning model. Dorian et al. (2017) explores the ability to detect humpback whale songs at different signal-to-noise ratio (SNR) levels. They worked with clean recordings of humpback whale songs from Sainte Marie Channel Island (located in North East Madagascar). The researchers wanted to test how well artificial intelligence can detect humpback whales through different types of noise. by combining the humpback whale songs with different types of background noise. Wind, rain, boats, and humpback whale song chorus were tested. They concluded that their convolutional neural network was better at detecting whales as the SNR increased for all of the different types of background noise.

2.2 Non-Humpback Whale Studies

Humpback whale songs may be relatively more complex, but this does not mean they are the only type of whale worth studying. Knowledge of whale presence and migration patterns are important for all whale species throughout the world. Some whale species are endangered, placing extra importance on this knowledge.

Garibbo et al. (2021) focused on the detection of a common fin whale call known as the "20Hz call". They use residual neural networks, a form of deep learning that focuses on image classification. Fin whale data was gathered from the LoVe observatory, offshore of Norway. Their model results coincided with the expected prevalence of fin whale calls during different parts of the year. Periods of frequent fin whale calls occurred during the breeding season, while periods of relatively infrequent fin whale calls occurred during periods of increased water traffic due to shipping.

Zhong et al. (2020) used neural networks to classify an endangered whale species, the beluga. Data was collected from the Cook Inlet beluga whale critical habitat, located in Alaska. A series of 2-second-long spectrogram snippets were collected. They use four differently architectured CNNs for spectrogram image classification. Their ensemble method featured great results of precision, accuracy, and ROC-AUC greater than 0.96, and a recall of 0.92.

Thomas et al. (2019) used different types of neural networks to detect and classify types of baleen whales. Data were collected during the fall months of 2015 and 2016 around the

Scotian Shelf. On their mel scale test, they obtained an accuracy score of 0.918, a precision score of 0.818, a recall score of 0.784, and an F-1 score of 0.801.

2.3 Non-Whale Studies

Passive acoustic monitoring extends beyond whales and other marine species. It is commonly used to detect other difficult to observe creatures, such as birds. Many of these studies implement audio transformation techniques and traditional machine learning models such as support vector machines and random forests, instead training neural networks on the spectrogram.

Ross & Allen (2013) make use of random forest models to identify passerine nocturnal flight calls among several other bird species. They emphasize the use of random forest models, as they don't require much tuning and are non-parametric. Random forest models were trained on spectro-temporal features, such as event duration, crest factor, and dominant frequency mean. Their random forest model helped eliminate several false detections, vastly reducing the manual annotation and identification time.

Salamon et al. (2016), focus on the classification and detection of different bird species on a couple different data sets. Their classification algorithms were effective at classifying bird species based on calls, but not as effective at the event detection. In particular, precision scores were poor for one data set, and the other data set suffered from many false positives due to frog species with audibly similar vocalizations in the data collection area.

2.4 Non-Animal

Sound detection and classification has applications outside of animal call detection entirely. Thambi et al. (2014) explores using sound detection to detect speech versus non-speech audio segments. They use spectral features such as spectral rolloff and spectral entropy as explanatory variables. Using audio from YouTube videos and recordings and a random forest algorithm, they obtained an overall accuracy score of 0.92 for their best model.

Phan et al. (2015) make use of random forests for classification for a variety of sounds, such as phone rings, door slams, and key jingles. They use Mel-frequency cepstral coefficients (MFCCs) to extract audio features on the individual sounds. With thousands of sound samples used for training, they were able to achieve high precision and recall scores. A similar workflow was performed by Alsouda et al. (2019). They used a variety of machine learning models, including support vector machines, k-nearest-neighbors, and random forests to differentiate audio data between sounds such as jackhammers, sirens, and street music. Their best models were able to achieve recall and precision scores above 0.90.

Sound detection and classification serves an important use in the field of security and fraud. Hamza et al. (2022) investigated using machine learning to detect deepfaked audio using 195,000 available speech samples. Using MFCCs and several different machine learning models, they were able to achieve above 99% accuracy on the classification of audio deepfakes, given an audio file.

Mohmmad and Sanampudi (2023) found that random forests and some convolutional neural networks were able to detect the illegal chopping of trees in actual forests. They achieved 75% accuracy with decision trees and 81% accuracy with their best neural network.

2.5 Gray Whale Study

The primary inspiration for this project involves the detection of gray whale songs by a group of students from California Polytechnic State University (Wong et al., 2025). Several thirty-minute wave files containing gray whale calls were provided to the gray whale project group by some marine science students (and Professor Schroth?). These recordings occurred over the course of a week in February 202(?) 3 to 4 miles offshore of Avila Bay. The students tried a few audio transformation functions including the Fast-Fourier Transform and MFCCs, and a few different traditional machine learning models (including an ensemble model). Their best model produced a recall score of 0.95 and precision score of 0.97. Application of their methods to humpback whale data did not produce good model metrics. We further discuss these methods, and build upon them in the next section.